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Needs Satisfaction Mediates the Relationships Between TPACK, Self-Efficacy and Teachers' Willingness to Integrate Generative AI Into Higher Education Practices

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Keywords: generative AI (GenAI) | higher education | self-determination theory (SDT) | self-efficacy | teachers' willingness | TPACK

ABSTRACT

The success of efforts to integrate generative AI (GenAI) into classrooms depends heavily on teachers' willingness to integrate the relevant technologies. However, few studies have examined the effect of perceived psychological needs satisfaction on teachers' willingness to integrate GenAI (WIAI). This study investigated the effect of perceived needs satisfaction according to self-determination theory (SDT) on the relationships between technological pedagogical content knowledge (TPACK), self-efficacy (SE) and WIAI among college teachers. Specifically, 1348 college teachers completed a comprehensive survey on the interconnectedness of TPACK, SE and SDT needs (autonomy, competence and relatedness) and their effects on WIAI. Structural equation modelling revealed that perceived SDT needs and SE directly affected WIAI. TPACK and SE significantly affected perceived SDT needs. The perceived SDT needs mediated the relationships of TPACK and SE with WIAI. These findings contribute to the literature of SDT, and provide practical recommendations for college administrators, tool developers and education researchers.

1 | Introduction

Generative AI (GenAI) has transformed various aspects of teaching and learning practices (Chiu 2024, 2025; Sun et al. 2024; Yin et al. 2024). As we embrace the digital age, the integration of GenAI stands poised to transform pedagogical landscapes. GenAI has the potential to enhance personalised learning, streamline administrative tasks, deliver real-time feedback and provide data-driven insights to improve educational outcomes (Wilder et al. 2024). For example, college

students increasingly use GenAI tools such as ChatGPT to assist with their learning and research and complete drafts and assignments (Rawas 2024); however, many instructors have not yet integrated these tools into their teaching practices. Educators' failure to consider the role of GenAI in instructional design may lead to a resource gap that reduces the efficacy of both their teaching and their students' learning. Teachers' willingness to adopt technologies will determine whether GenAI can be integrated with classroom practices (Sun et al. 2024; Wang, Chen, et al. 2024). This willingness

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to integrate GenAI (WIAI) may be understood in the context of self-determination theory (SDT), which emphasises individual motivation through the satisfaction of needs, namely the needs for competence, autonomy and relatedness (Chiu 2022; Ryan and Deci 2000). Studies on the integration of information technology with teaching have drawn on SDT (Chang et al. 2017; Nikou and Economides 2017; Şahin and Şahin 2022). When educators' psychological needs are met, they are increasingly likely to feel empowered and motivated to integrate GenAI into their teaching practices as a means of improvement (Chiu 2022; Chiu et al. 2024).

Recent research examining the factors influencing teachers' WIAI has focused primarily on the technology acceptance model (TAM) and teachers' perceptions of GenAI in educational settings (Mutammimah et al. 2024; Wang, Chen, et al. 2024). However, the TAM is predominantly technology-focused and neglects other factors that may explain teachers' reluctance to include information and communications technologies in their educational practices (Kreijns et al. 2013). Although SDT has been applied in efforts to integrate technology with teaching (Chang et al. 2017; Nikou and Economides 2017; Şahin and Şahin 2022), to the best of the authors' knowledge, few studies have explored how the satisfaction of psychological needs shapes teachers' WIAI. Studies on GenAI integration have primarily focused on pre-service teachers (Sun et al. 2024; Wang, Ruan, et al. 2024; Yin et al. 2024) or K-12 educators (An et al. 2023); few have focused on instructors in higher education.

Despite growing interest in GenAI within higher education, research on teachers' WIAI from the SDT perspective remains limited, hindering our understanding of how to integrate GenAI in higher education. This study aims to address these gaps by investigating how the satisfaction of SDT needs might influence the relationships between technological pedagogical content knowledge (TPACK), self-efficacy (SE) and WIAI. The findings of this study could offer valuable insights into how SDT can inform the integration of GenAI tools in higher education. Furthermore, the study may provide practical implications for college administrators, educational GenAI tool developers and education researchers, by suggesting ways to enhance teachers' WIAI through better support for their SDT needs and professional development.

2 | Literature Review

2.1 | SDT And WIAI

SDT provides a theoretical framework for analysing the factors influencing teachers' WIAI. SDT asserts that three fundamental psychological needs—competence, relatedness and autonomy—have a crucial influence on intrinsic motivation and engagement in various contexts (Ryan and Deci 2017, 2020). These SDT needs are related to the integration of GenAI tools, such as ChatGPT. Chiu (2024) developed a classification tool based on SDT to support and encourage teachers in leveraging GenAI more effectively for their own teaching. In addition, a study showed that SDT needs are significant predictors of learners' behavioural intentions and actual use of GenAI tools; and this study introduced SDT motivation as a moderator to reveal more insights

that SDT significantly moderates the impact of technology use conditions on behavioural intentions and actual use (Zheng et al. 2024). Individuals who experience autonomy, which enables them to control their thoughts and behaviours, are more likely than others to develop a positive attitude (Deci et al. 2017; Ryan and Deci 2017). Competence encourages favourable attitudes towards new technologies by boosting teachers' self-confidence and sense of understanding (Chiu et al. 2024; Yang et al. 2025). For example, Dahri et al. (2024) determined that personal competency significantly and positively influenced the acceptance of GenAI tools, such as ChatGPT. When their need for relatedness is met, teachers may hold positive attitudes and feel a sense of belonging with peers who use similar AI technologies (Ryan and Deci 2017; Şahin and Şahin 2022).

The satisfaction of these three psychological needs has been found to play a pivotal role in shaping teachers' willingness to integrate new technologies into their teaching practices (Chang et al. 2017; Nikou and Economides 2017; Şahin and Şahin 2022). In a study involving the integration of TAM and SDT, Nikou and Economides (2017) demonstrated that the satisfaction of SDT needs was positively correlated with the perceived ease of use and usefulness of technology and thus influenced the adoption of that technology. In a European study, Bergdahl et al. (2023) found that fulfilling psychological SDT needs was associated with greater AI positivity and reduced negativity; they further found that autonomy and relatedness enhanced these associations. Additionally, the findings of Yang et al. (2025) are similar to these results, indicating that SDT needs satisfaction can significantly buffer the weakening effect of teachers' negative emotions on WIAI. Dahri et al. (2024) found that personal competency, social influence and positive attitudes significantly influenced the acceptance of ChatGPT. Ateş and Gündüzalp (2024) highlighted the significant impact of perceived relatedness on teachers' willingness to adopt robotics in STEM education. Accordingly, we hypothesise that the satisfaction of each of the three SDT needs promotes teachers' WIAI (H1).

2.2 | How TPACK Affects Needs Satisfaction and WIAI

TPACK has emerged as a significant framework in educational technology, highlighting educators' ability to integrate technology into instructional design effectively (Mishra et al. 2023; Shulman 1986). It has also been recognised as a key factor influencing teachers' willingness to integrate GenAI into their practices (An et al. 2023; Yang et al. 2025; Sun et al. 2024). Building upon Shulman's foundational concept of pedagogical content knowledge (PCK), TPACK synthesises technological, pedagogical and content knowledge, thereby giving rise to specific knowledge domains such as technological content knowledge (TCK), pedagogical content knowledge (PCK) and technological pedagogical knowledge (TPK) (Koehler et al. 2014). Studies have shown that educators who possess higher levels of TPACK are more confident and willing to include innovative technologies in their teaching practices (Koehler et al. 2014; Niess 2011; Yang et al. 2021). For example, Sun et al. (2024) found that TPACK had a direct impact on the WIAI of pre-service STEM teachers. An et al. (2023) reported that TPACK significantly predicted K-12

language teachers' behavioural intention to use AI in their teaching. Furthermore, Durham (2024) conducted a survey of pre-service teachers and found through qualitative analysis that they purposefully used the TPACK framework to reflect and choose digital tools to use that would support students' language development and leverage students' interests. Accordingly, TPACK is likely to be related to WIAI (H2).

In the field of educational technology, TPACK is a key framework used to determine how proficiently educators integrate technology into their teaching practices (Karataş and Ataç 2024; Mishra et al. 2023; Shulman 1986). For example, Mishra et al. (2023) provided a groundbreaking expansion of the TPACK model by incorporating GenAI considerations and redefining Contextual Knowledge (XK). Thyssen et al. (2023) proposed extending the TPACK model to the DPACK model by incorporating digital cultural transformation. Al-Adwan, Meet, Kala, et al. (2024) claimed that TPACK was identified as a critical factor in enabling the effective integration of technology into teaching practices. Expanding on Falloon's (2020) TPACK framework, Chiu et al. (2024) found that the fulfilment of the basic psychological needs outlined in SDT significantly enhanced teachers' digital competence. Chen et al. (2024) found that the technological knowledge (TK) and pedagogical knowledge (PK) within the TPACK framework may indirectly enhance teachers' autonomy-supportive behaviours in online teaching by improving their digital competence. Koehler et al. (2014) and Niess (2011) reported that teachers with a high level of TPACK preferred to use technology in their teaching. In summary, TPACK appears to influence teachers' WIAI by satisfying their need for competence.

Regarding the need for relatedness, TPACK enhances teachers' sense of relatedness by fostering a sense of belonging and connection within the teaching community (Chiu 2022; Chiu et al. 2024). In the study by Yang et al. (2025), the results revealed a significant positive correlation between TPACK and the satisfaction of perceived relatedness needs. The integration of technology, pedagogy and content allows teachers to collaborate and share their best practices and thus strengthens their sense of relatedness with their peers, schools and communities (Fathali and Okada 2018).

Regarding autonomy, Wang, Chen, et al. (2024) demonstrated that TPACK, teacher agency, and teacher autonomy are positively associated with teachers' behavioural intentions to design AI-assisted learning. Roumbanis Viberg et al. (2021) claimed that autonomy is a core factor for Teacher Educators to achieve professional agency and professional identity in a digitalised society. A qualitative study found that developing EFL teachers' TPK and autonomy is essential for the effective implementation of computer-assisted education, enabling them to adapt to technological and pedagogical challenges while addressing emergent educational needs (Tafazoli and Meihami 2023). Chen et al. (2024) studied how TPACK influences teachers' sense of autonomy and found that it is an important contributor to a supportive and autonomous teaching environment. According to their research, teachers with a higher level of TPACK have a greater capacity for cognitive empathy, which enables them to tailor their integration of technology according to their pedagogical goals and students'

needs. This tailored integration of technology, guided by TPACK, empowers teachers, enhancing their autonomy regarding their classroom practices.

Accordingly, we hypothesise that TPACK has a positive impact on the satisfaction of the three SDT needs (competence, relatedness, autonomy) (H3).

2.3 | How SE Affects Needs Satisfaction and WIAI

SE regarding GenAI is another important factor that affects teachers' WIAI in their teaching practices (Strzelecki et al. 2024; Sun et al. 2024). SE is a crucial factor in the uptake of new technologies (Montag et al. 2023; Strzelecki et al. 2024; Wang and Zhao 2021). For example, a study conducted by Şahin et al. (2021) by extending TAM showed that self-efficacy significantly affects the intention to use information technologies. Additionally, studies have shown that SE influences both the perceived ease of use and perceived usefulness of AI (Al-Adwan, Meet, Anand, et al. 2024; Dehghani and Mashhadi 2024), which in turn directly impacts teachers' willingness to use AI (Mutammimah et al. 2024; Wangdi and Rigdel 2024; Wijaya et al. 2024). For instance, Wang, Ruan, et al. (2024) identified technology SE as an important determinant of pre-service teachers' performance expectancy and their behavioural intentions. Sun et al. (2024) found that SE has a direct impact on pre-service STEM teachers' WIAI. In summary, SE is likely to be related to WIAI (H4).

SE is a key psychological determinant of an individual's motivation and behaviour (Bandura 1982). In this context, it refers to teachers' belief in their ability to effectively implement technology into their instructional practices (i.e., competence). Al-Adwan, Meet, Anand, et al. (2024); Al-Adwan, Meet, Kala, et al. (2024) found that SE had a statistically significant effect on the intention to continuously use technology. Guo et al. (2025) identified self-efficacy as a key factor influencing teachers' acceptance of AI. Teachers with higher SE are more likely to feel competent when integrating new technologies: their strong belief in their capabilities enables trust, a crucial mediator between SE and their WIAI (Al-Adwan, Meet, Anand, et al. 2024; Al-Adwan, Meet, Kala, et al. 2024; Montag et al. 2023; Wang, Ruan, et al. 2024).

Relatedness is also essential to SE and GenAI integration. Wang and Chuang (2024) recently developed an AI SE (AISE) scale to explore the relatedness between individuals and AI-based technologies, thus providing a tool to help educators and practitioners assess individuals' AISE and understand the behaviours associated with AI use. This work further emphasises the importance of relatedness in AI integration. Other research has found a significant relationship between college students' SE and use of AI tools, such that students with higher SE are more likely to use AI to solve daily problems and academic tasks (Rodríguez-Ruiz et al. 2024). For example, Adigun et al. (2025) found that technological SE had a significant positive and indirect effect on pre-service teachers' behavioural intention to adopt and use AI for inclusive education teaching, mediated by performance expectancy and social influence. Even in a normal educational context, there are significant differences in relatedness and collaboration among teachers with different levels of SE (Huang

et al. 2024). Again, these findings suggest that relatedness plays a key role in the relationship between the SE and WIAI of college teachers.

Finally, autonomy plays an important part in the relationship between SE and WIAI. In digital contexts, SE can directly influence teachers' autonomous motivation (Moreira-Fontán et al. 2019). This autonomy is further reinforced when teachers perceive that their independence is supported by the learning environment, which in turn boosts their motivation (Duchatelet and Donche 2019). Autonomy over their teaching practices empowers teachers to experiment with AI tools and adjust the tools to meet their students' needs, ultimately increasing the likelihood that teachers will include GenAI in their teaching practice; accordingly, their WIAI is increased.

Accordingly, we hypothesise that SE contributes to the satisfaction of the three SDT needs (H5), and that needs satisfaction mediates the effects of TPACK and SE on teachers' WIAI (H6, H7).

2.4 | Research Gaps

Despite the existing research on technology integration in teacher education, several significant gaps remain. First, while TPACK has been widely recognised as a determinant of technology adoption (Celik 2023; Mishra and Koehler 2006; Wang, Ruan, et al. 2024), the role of psychological dimensions such as SE and needs satisfaction in mediating this relationship has not been thoroughly explored. In addition, most studies have emphasised cognitive and technical factors, such as the framework of TPACK and TAM, neglecting the psychological and affective states that influence teachers' WIAI.

Second, existing research has mainly focused on pre-service teachers or K-12 educators (Backfisch et al. 2024; Kong and Yang 2024; Rizvi et al. 2023; Yang et al. 2024), and there is a clear research gap on how higher education faculty, who shoulder the dual responsibilities of teaching and academic research, can engage with GenAI technologies. Teacher groups work in different institutions and levels, not just K-12 education, and this research gap limits the generalisation of their findings to higher education settings.

Finally, while the satisfaction of psychological needs outlined in SDT is considered key to motivating technology adoption (Xia et al. 2022, 2023; Ryan and Deci 2017), there is little empirical evidence on how these needs interact with TPACK and SE to shape teachers' WIAI. Therefore, further empirical research is needed to understand the relationships and mechanisms of these needs.

2.5 | This Study and Hypotheses

The objective of this study was to determine the factors that influence teachers' WIAI in their teaching practice. Specifically, we investigated how teachers' TPACK and SE affect WIAI and how these relationships are mediated by the satisfaction of SDT needs (competence, relatedness, autonomy). The proposed

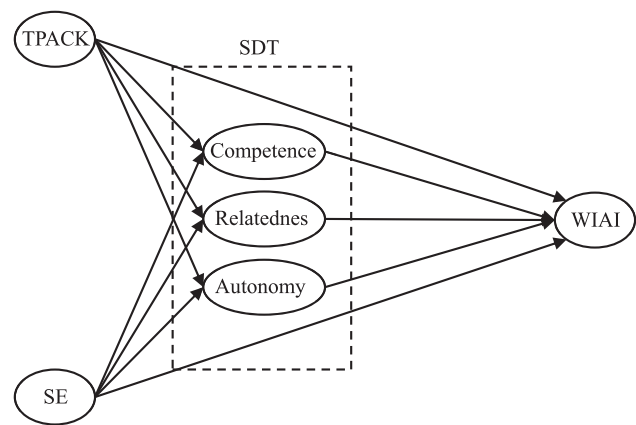


FIGURE 1 | Research model.

research model is illustrated in Figure 1, and we propose the following hypotheses:

- H1: Satisfaction of SDT needs promotes WIAI.
- H2: TPACK is related to WIAI.
- H3: TPACK promotes the satisfaction of the three SDT needs (competence, relatedness, autonomy).
- H4: SE is related to WIAI.
- H5: SE contributes to the satisfaction of the three SDT needs.
- H6: Satisfaction of the SDT needs mediates the effects of TPACK on WIAI.
- H7: Satisfaction of the SDT needs mediates the effects of SE on WIAI.

3 | Method

3.1 | Participants and Data Collection

This study employed a stratified purposive sampling approach to select 1348 college teachers from 28 cities or provinces, ensuring diverse representation based on geographical location and institutional characteristics. All participants provided informed consent before taking part in the survey. Data were collected online in July 2024 via a nationwide survey organised by institutes affiliated with the Ministry of Education's Centre for Scientific Research and Development in Higher Education. According to Soper (2020), for structural equation modelling (SEM), a minimum of 200 participants was required to detect a medium effect size ($f^2 = 0.15$) with $\alpha = 0.05$ and power = 0.95, considering a model with six latent constructs and 19 observed variables. Additionally, with a participant-to-parameter ratio exceeding the recommended 10:1 threshold ($19 \times 10 = 190 < 1348$), our sample size was therefore sufficient to validate the proposed structural equation model (Kline 2016).

The participants were drawn from multiple academic disciplines, ranging from the Humanities and social sciences to science, technology, engineering, agriculture and medicine

(Science and applied disciplines). The survey was administered online, allowing us to easily monitor the response rates. To guarantee data completeness, all of the survey items were compulsory, ensuring that the participants responded to each question before submission. Detailed instructions were provided during the distribution to ensure clarity and accuracy in responses. As a result, our dataset was complete (i.e., no missing values). The demographic characteristics of the participants are given in Table 1.

3.2 | Questionnaires

The survey instrument measured six key variables: TPACK, SE, the three components of SDT needs and WIAI. For each variable, the participants responded to three or four items using a 5-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree). A comprehensive list of these items can be found in the Appendix.

Perceived TPACK was measured using three items adapted from Sun et al. (2024); the reliability of the original scale was $\alpha = 0.803$. An example item is 'I can integrate professional knowledge, technology and teaching knowledge appropriately to design student-centred learning activities'.

Perceived SE was measured using three items adapted from Bandura (1977); the reliability of the original scale was $\alpha = 0.963$. An example item is 'I believe that I can keep up with the latest developments in GenAI technology in the field of education'.

The three perceived SDT needs, namely competence, relatedness and autonomy, were evaluated using items derived from Ryan and Deci (2000). Perceived competence was measured using three items (e.g., 'I always keep up with the latest technology'),

which had a reliability of $\alpha = 0.883$. Perceived relatedness was also measured using three items (e.g., 'The use of GenAI in the classroom is compatible with my regular teaching methods'), which had a reliability of $\alpha = 0.937$. Perceived autonomy was assessed using four items (e.g., 'In using GenAI for teaching, I have the freedom to design the teaching content'), which had a reliability of $\alpha = 0.917$.

WIAI was assessed using four items adapted from Sun et al. (2024), which had an original reliability of $\alpha = 0.806$. A sample item is 'In the future, I plan to use GenAI technology as a regular teaching method'.

3.3 | Data Analysis

Data analysis was conducted using Mplus v8.3 and SPSS v26.0. First, the composite reliability of each subscale was calculated to evaluate the internal consistency; the corresponding Cronbach's α values are reported in the Questionnaires section. Second, confirmatory factor analysis (CFA) was performed to evaluate construct validity. Finally, a path analysis was performed to identify relationships between the variables. Mediation effects were tested using the bootstrapping method.

4 | Results

4.1 | Descriptive Statistics and Reliability and Validity Testing

Table 2 lists the descriptive statistics for each factor. The skewness ranged from -0.893 to -0.134 and the kurtosis from 0.322 to 1.559 . As all values fell within the acceptable range of ± 2 , as established by Kline (2023), the data were generally normally distributed and compatible with subsequent analyses.

We assessed construct validity using CFA. The results indicated a good fit of the measurement model to the data ($\chi^2 = 811.030$, $df = 137$, $CFI = 0.975$, $TLI = 0.968$, $RMSEA = 0.060$, $SRMR = 0.033$) (Hu and Bentler 1999). The composite reliability (CR) was computed for each dimension, and all CR values exceeded the recommended threshold of 0.70 , which indicated good internal consistency (Table 3) (Hair Jr et al. 2020). All of the indicators had standardised factor loadings above 0.60 and were significantly loaded ($p < 0.001$) on each latent variable. For all variables, the average variance extracted (AVE) exceeded 0.50 ;

TABLE 1 | Demographic information of participants.

Variable	Frequency	Percentages
Gender		
Male	582	43.2
Female	766	56.8
Subject		
Science and applied disciplines	936	69.4
Humanities and social sciences	412	30.6
Age		
Under 30 years	140	10.4
Between 30 and 40 years	473	35.1
Between 40 and 50 years	518	38.4
Over 50 years	217	16.1

TABLE 2 | Descriptive statistics of variables.

Variables	M	SD	Skewness	Kurtosis
1 TPACK	3.861	0.819	-0.893	1.559
2 Competence	3.693	0.813	-0.524	0.740
3 Relatedness	3.811	0.825	-0.622	0.953
4 Autonomy	3.494	0.815	-0.134	0.322
5 SE	3.988	0.783	-0.746	1.245
6 WIAI	3.714	0.759	-0.297	0.609

therefore, each subscale had strong convergent validity. The square root of the AVE for each variable was greater than its correlations with other variables, which confirmed the discriminant validity (Fornell and Larcker 1981; Pallant 2020).

4.2 | Mediating Effect Analysis

We constructed a path model to further evaluate the relationships between the variables. The SEM results indicated that the model depicted in Figure 1 provided a good fit for the data ($\chi^2=936.220$, $df=140$, $CFI=0.970$, $TLI=0.963$, $RMSEA=0.065$, $SRMR=0.042$). Figure 2, Tables 4 and 5 present the model path coefficients, standardised direct, indirect and total effects identified among the variables.

4.3 | Relationship Between Perceived Needs Satisfaction and WIAI

The perceived satisfaction of each need had a significant positive effect on teachers' WIAI (H1). Specifically, perceived competence ($\beta=0.166$, $p<0.01$), perceived relatedness ($\beta=0.295$, $p<0.001$) and perceived autonomy ($\beta=0.272$, $p<0.001$) were all significant predictors of WIAI.

4.4 | Relationship Between TPACK and Perceived Needs Satisfaction

For teachers, TPACK significantly and positively predicted the satisfaction of all three perceived SDT needs (H3). Specifically, TPACK had a significant positive impact on perceived competence ($\beta=0.324$, $p<0.001$), perceived relatedness ($\beta=0.226$, $p<0.001$) and perceived autonomy ($\beta=0.090$, $p<0.05$).

4.5 | Relationship Between SE and Perceived Needs Satisfaction

SE also significantly and positively predicted the perceived satisfaction of each of the three SDT needs (H5). SE had significant effects on perceived competence ($\beta=0.588$, $p<0.001$), perceived relatedness ($\beta=0.633$, $p<0.001$) and perceived autonomy ($\beta=0.124$, $p<0.01$).

4.6 | Mediating Role of Perceived Needs Satisfaction

To determine the mediating role of needs satisfaction, we used a bootstrapping approach with 1000 random samples and

TABLE 3 | Reliability and validity.

Variables	CR	Factor loadings	AVE	1	2	3	4	5	6
1 TPACK	0.931	0.860, 0.946, 0.906	0.818	0.905					
2 Competence	0.889	0.785, 0.865, 0.905	0.728	0.691	0.854				
3 Relatedness	0.938	0.922, 0.943, 0.875	0.836	0.622	0.773	0.914			
4 Autonomy	0.924	0.915, 0.945, 0.876	0.752	0.487	0.622	0.644	0.867		
5 SE	0.963	0.933, 0.955, 0.951	0.896	0.636	0.786	0.770	0.817	0.947	
6 WIAI	0.936	0.756, 0.924, 0.910, 0.869	0.785	0.512	0.689	0.709	0.817	0.729	0.886

Note: $N=1348$. All correlations are significant ($p<0.001$). Bold values represent the square root of the AVE for each variable.

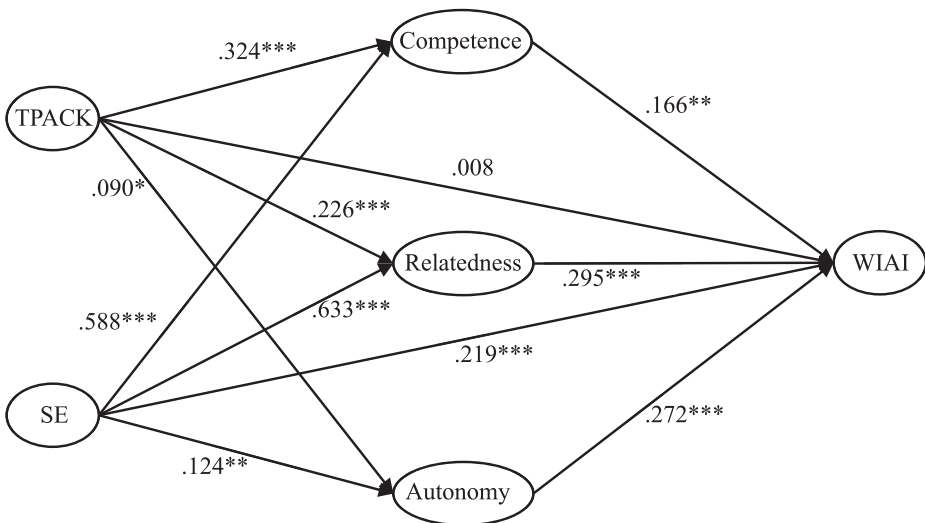


FIGURE 2 | Model path coefficients.

TABLE 4 | Standardised total, indirect and direct effects of TPACK on WIAI.

Path		Standardised Estimate	S.E.	95% CI (Lower)	95% CI (Upper)
Total effects from TPACK to WIAI	Total	0.153***	0.037	0.093	0.215
	Total indirect	0.145***	0.025	0.108	0.191
Indirect effects	TPACK→Competence→WIAI	0.054**	0.019	0.025	0.085
	TPACK→Relatedness→WIAI	0.067***	0.015	0.042	0.095
	TPACK→Autonomy→WIAI	0.024*	0.011	0.007	0.043
Direct effects	TPACK→WIAI	0.008	0.038	−0.053	0.069
	TPACK→Competence	0.324***	0.041	0.256	0.386
	TPACK→Relatedness	0.226***	0.044	0.163	0.307
	TPACK→Autonomy	0.090*	0.040	0.026	0.157
	Competence→WIAI	0.166**	0.053	0.069	0.246
	Relatedness→WIAI	0.295***	0.045	0.219	0.363
	Autonomy→WIAI	0.272***	0.026	0.228	0.312

* $p < 0.05$.** $p < 0.01$.*** $p < 0.001$.**TABLE 5** | Standardised total, indirect and direct effects of SE on WIAI.

Path		Standardised Estimate	S.E.	95% CI (Lower)	95% CI (Upper)
Total effects from SE to WIAI	Total	0.537***	0.035	0.471	0.590
	Total indirect	0.318***	0.048	0.241	0.395
Indirect effects	SE→Competence→WIAI	0.098**	0.032	0.041	0.148
	SE→Relatedness→WIAI	0.187***	0.034	0.130	0.239
	SE→Autonomy→WIAI	0.034**	0.012	0.016	0.055
Direct effects	SE→WIAI	0.219***	0.058	0.128	0.318
	SE→Competence	0.588***	0.039	0.522	0.649
	SE→Relatedness	0.633***	0.041	0.561	0.693
	SE→Autonomy	0.124**	0.039	0.062	0.191
	Competence→WIAI	0.166**	0.053	0.069	0.246
	Relatedness→WIAI	0.295***	0.045	0.219	0.363
	Autonomy→WIAI	0.272***	0.026	0.228	0.312

* $p < 0.05$.** $p < 0.01$.*** $p < 0.001$.

calculated 95% confidence intervals (CI) for the indirect effects. This method, which avoids distributional assumptions, allowed us to robustly test the mediation pathways.

First, the total effect of TPACK on WIAI was 0.153 ($p < 0.001$). The indirect effects of TPACK mediated through perceived competence ($\beta = 0.054$, $p < 0.01$), perceived relatedness ($\beta = 0.067$, $p < 0.001$) and perceived autonomy ($\beta = 0.024$, $p < 0.05$) were all significant, whereas the direct effect of TPACK on WIAI (H2)

was not significant ($\beta = 0.008$, $p > 0.05$). The 95% CIs for the indirect effects did not include zero, confirming that the perceived satisfaction of SDT needs significantly mediated the relationship between TPACK and WIAI (H6).

Second, the total effect of SE on WIAI was 0.537 ($p < 0.001$). The indirect effects of SE mediated through perceived competence ($\beta = 0.098$, $p < 0.01$), perceived relatedness ($\beta = 0.187$, $p < 0.001$) and perceived autonomy ($\beta = 0.034$, $p < 0.01$) were all

significant, and the direct effect of SE on WIAI was significant ($\beta = 0.219$, $p < 0.001$), indicating partial mediation. The 95% CIs for these indirect effects did not include zero, suggesting that SE influenced WIAI both directly and indirectly via the perceived satisfaction of SDT needs (H4, H7).

Overall, these results confirm that the perceived satisfaction of the three SDT needs (competence, relatedness and autonomy) mediates the relationships of TPACK and SE with WIAI.

5 | Discussion and Implications

We evaluated the mediating role of teachers' perceived SDT needs satisfaction in the relationships of their SE and TPACK with their WIAI in the context of higher education. Accordingly, this important study makes three empirical contributions and two theoretical contributions to the literature. Additionally, we offer several practical suggestions for higher education leadership, educational GenAI developers and education researchers.

5.1 | Empirical Implications

First, we found a significant direct effect of TPACK on perceived SDT needs satisfaction (H3); however, the direct effect of TPACK on WIAI was nonsignificant (H2). These findings suggest that teachers with a higher TPACK are more likely to achieve satisfaction of their perceived SDT needs, consistent with most SDT-related research (Chen et al. 2024; Rienties et al. 2022). This indicates that TPACK serves as a foundational skill set that indirectly enhances teachers' intrinsic motivation to integrate GenAI tools by addressing their psychological needs satisfaction. Chen et al. (2024) discovered that digital competence enhanced teachers' ability to provide their autonomy support in online teaching, and that this relationship was mediated by cognitive empathy. In other words, teachers with higher TPACK are more likely to achieve satisfaction of their SDT needs in their teaching practices (Chen et al. 2024; Rienties et al. 2022). Our findings that TPACK was less likely to influence teachers' WIAI contradict the findings of Sun et al. (2024). One plausible explanation for this discrepancy is that GenAI tools such as ChatGPT are user-friendly and present minimal technical challenges; accordingly, they can be used effectively even by teachers with low levels of TPACK.

Second, we found that teachers' SE positively influenced both the perceived satisfaction of the three SDT needs (H5) and WIAI (H4). These findings are generally consistent with the body of SDT-related field research (Chiu 2022; Montag et al. 2023; Wang and Zhao 2021). For example, Montag et al. (2023) found that technology SE enhanced individuals' trust in and attitudes towards AI. Sun et al. (2024) found that SE directly affected pre-service STEM teachers' WIAI. These findings suggest that teachers with higher SE are more likely to have their SDT needs satisfied and thus are more inclined to adopt GenAI in their classrooms, possibly because they feel that they have the competence to use GenAI. Additionally, the findings reveal that SE had a greater influence on teachers' WIAI than TPACK, which suggests that teachers' perceived psychological state and confidence are crucial determinants of their technology adoption.

Specifically, when teachers believe that they can use AI effectively, they will be more likely to embrace new technologies even if they lack advanced technical and pedagogical knowledge. Therefore, when promoting GenAI use in higher education, enhancing teachers' SE may be more important than simply increasing their TPACK.

Third, we found that the satisfaction of each of the three needs positively affected teachers' WIAI (H1) and also mediated the effects of TPACK and SE on WIAI (H6, H7). These results are consistent with those of most prior SDT-related studies in the field of educational technology (Barrable and Lakin 2020; Chang et al. 2017; Nikou and Economides 2017; Qiu and Li 2024; Roca and Gagné 2008), further supporting the hypothesis that satisfying the three SDT needs can promote teachers' willingness to use a new technology. Our findings regarding the mediating effects of satisfaction of SDT needs indicate that supporting teachers' needs for autonomy, competence and relatedness is crucial for promoting their WIAI in their teaching practices. One plausible explanation for this relationship is that willingness refers to an individual's readiness or inclination to engage in a specific action, which is influenced by their intrinsic motivation (Ajzen 1991). For teachers, satisfying their SDT needs may enhance their intrinsic motivation. When their psychological needs are met, teachers feel capable, autonomous and supported, and these feelings increase their WIAI in their teaching practices. Additionally, AI technology primarily affects cognition rather than emotions, according to the theory of mind in AI (Langley et al. 2022). According to this theory, social information can be processed regardless of whether its source is AI or another person (Lee 2004). In other words, GenAI cannot directly influence teachers' WIAI in the classroom; instead, teachers' psychological needs should be addressed.

5.2 | Theoretical Implications

This study provides a new theoretical perspective on the integration of technology with higher education by identifying SE and needs satisfaction as key factors in this process. For instance, previous theoretical approaches such as the TAM and TPACK have primarily focused on cognitive aspects, including users' technological knowledge (TK), content knowledge (CK) and pedagogical knowledge (PK). The findings of most research on technology integration in education suggest that TK (e.g., computer SE and TPACK) influences teachers' intention to use AI technology in the classroom (An et al. 2023; Sun et al. 2024). However, we found that supporting teachers' needs for autonomy and satisfaction of SDT needs, but not TPACK, directly influenced teachers' WIAI. This finding may encourage researchers to re-examine existing theories and models by exploring the relationships between SE, TPACK and WIAI from the perspective of SDT.

Second, little is known about psychological needs relating to the integration of GenAI in teacher education. This study on GenAI in educational technology contributes to research on SDT by demonstrating how needs satisfaction can explain teachers' WIAI in higher education settings. Most SDT-based research on digital learning environments, such as self-regulated learning (SRL) and games, has focused on students' learning (Chiu 2021;

Bedenlier et al. 2020; Xie and Ke 2011). We take a different perspective by emphasising how needs satisfaction can affect teachers' educational practices.

5.3 | Practical Implications

Our findings offer three practical implications for higher education leaders, educational GenAI developers and education researchers. First, our findings suggest that university leaders should provide education students with professional development training that satisfies their SE and perceived SDT needs, as these factors significantly influence teachers' WIAI in the classroom. For instance, university leaders can offer comprehensive training sessions that emphasise practical applications of GenAI in teaching. As Stalder (2018) notes, the rise of digital technology is reshaping knowledge production and dissemination, requiring educators to adapt their methods and integrate new tools effectively. Therefore, allowing teachers to help select GenAI tools that are tailored to their specific subjects and teaching styles would foster their autonomy, and their sense of competence can be increased by clarifying the teaching tasks that will involve GenAI. To increase relatedness, collaborative learning communities can be established to allow teachers to share their experiences, challenges and successes with GenAI. Addressing teachers' SE and psychological needs will enhance their WIAI.

Second, our findings can inform the developers of educational GenAI tools. According to the theory of mind in AI, interactions between humans and AI occur predominantly at the cognitive level, rather than at the psychological and emotional levels (Langley et al. 2022; Lee 2004; Lu et al. 2019). Therefore, the developers of educational AI tools should direct their attention towards teachers' psychological needs and emotions. For instance, developers should conduct comprehensive surveys to determine teachers' needs and attitudes. Development should focus not only on student learning but also on teachers' experiences and emotions when applying AI in their teaching practices. By addressing teachers' psychological needs, developers can enhance teachers' WIAI in the classroom. Furthermore, educators are increasingly integrating technology across the curriculum, that there is a fairly low level of TPACK adoption, and that conceptions of TPACK vary greatly (Voithofer and Nelson 2021). According to our findings (H3), TPACK has a significant positive impact on the demand for SDT. Therefore, educational GenAI tool developers should also consider the TPACK level of teachers and develop a variety of educational technology tools to meet their SDT needs.

Third, we urge researchers in the field of higher education AI to focus on the psychological needs of teachers, including their emotions, feelings and personalities. Research on the applications of technology in education has focused mainly on the cognitive and behavioural aspects of the TAM (Kreijns et al. 2013; Sun et al. 2024; Wang, Ruan, et al. 2024); in comparison, the influence of one's psychological state on the use of GenAI has received little attention (Xia et al. 2023), especially in the context of higher education teacher training. According to recent studies, teachers' emotions and perceived psychological states strongly influence their willingness to use technology (Şahin

and Şahin 2022; Yang et al. 2025; Yin et al. 2024). Our findings further support this perspective in the context of technology-enhanced education. Future research should focus on the psychological factors affecting university teachers' adoption of technology.

6 | Conclusion

This study examined how the perceived psychological needs influence the relationships between TPACK, SE and WIAI among college teachers. The results revealed that the satisfaction of SDT needs and SE directly influenced WIAI, while TPACK and SE significantly contributed to the satisfaction of SDT needs. Furthermore, the satisfaction of SDT needs mediated the effects of TPACK and SE on WIAI, highlighting their central role in fostering teachers' willingness to integrate AI into instruction. These findings contribute to the literature on SDT by demonstrating how the perceived needs satisfaction influences teachers' AI adoption. Additionally, this study highlights the need to support teachers' SDT needs in efforts to promote AI integration in higher education.

This study has three main limitations. First, as our proposed model is evaluated using SEM, further research using different methodologies, such as longitudinal, experimental, or intervention-based studies, is needed to corroborate our findings. We recommend that researchers conduct intervention-based or experimental studies to further assess whether SDT needs satisfaction facilitates teachers' WIAI. Second, our questionnaire was limited to college teachers. However, variations in regional and age demographics may yield different results. In future studies, data from diverse regions and age groups could be analysed and compared to enhance the generalisability of the findings. Lastly, although our study used a quantitative approach (i.e., questionnaire survey), we were unable to conduct a comprehensive analysis of the relationships between perceived needs satisfaction, SE and WIAI. Qualitative methodologies could be used in future studies to enable a more nuanced and in-depth exploration of these relationships.

Author Contributions

Qi Xia: conceptualisation, methodology, formal analysis, original draft. **Yiming Yang:** methodology, original draft. **Xiaoqing Weng:** reviewing and editing, methodology, formal analysis, validation. **Wing Kin Cheng:** reviewing – editing, data curation, validation. **Thomas K. F. Chiu:** conceptualisation, methodology, reviewing and editing.

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Ethics Statement

This study obtained ethical approval from the Survey and Behavioural Research Ethics Committee of the Chinese University of Hong Kong (SBRE-21-0488) before conducting the survey.

Consent

We obtained informed consent from all participants before they took part in the survey.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Appendix A

Technological Pedagogical and Content Knowledge (TPACK)

I can use some information technology to conduct formative and summative evaluations on students.

I can use appropriate digital technology to help students build professional knowledge according to the curriculum standards.

I can integrate professional knowledge, technology, and teaching knowledge appropriately to design student-centred learning activities.

Self-Efficacy

I believe that I can keep up with the latest developments in GenAI technology in the field of education.

I have the ability to use GenAI technology to achieve teaching goals.

I can choose appropriate GenAI technology based on future teaching content.

Self-Determination Theory (SDT) Needs Satisfaction

Competence.

GenAI makes me feel good at teaching.

GenAI helps me feel capable of effectively teaching students in class.

GenAI boosts my confidence in delivering instruction.

Relatedness.

The use of GenAI in the classroom is compatible with my regular teaching methods.

GenAI technology can cater to the teaching methods I currently use.

The content generated by GenAI is helpful for me to learn relevant knowledge.

Autonomy.

In using GenAI for teaching, I have the freedom to design the teaching content.

In school, I have a say in which GenAI technologies are used.

My teaching philosophy is freely expressed when applying GenAI to teaching.

Willingness to Integrate GenAI (WIAI)

In the future, I plan to use GenAI technology as a regular teaching method.

I am confident that I can effectively deal with any GenAI-related situations that suddenly occur in teaching.

I am always happy to apply new AI technology to learning activities of subject content.

I look forward to using GenAI technology to support my teaching work.